

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA1220	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Industry Problem solving task
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:	Sanjay Kumar P	Reg. No.:	192565073
Section / Batch:	Slot B	Date:	17-08-2026

Instructions to Students

- Identify the relevant concepts of cache hierarchy, SRAM, DRAM, MRAM, NAND Flash, RRAM, memory latency, bandwidth, and virtual memory paging.
- Analyze the memory-access requirements of smartphones, cloud servers, autonomous vehicles, edge AI devices, and gaming consoles, considering latency, bandwidth, capacity, and power consumption.
- Apply suitable memory-hierarchy techniques such as cache optimization, appropriate memory technology selection, data locality, paging optimization, and hierarchical memory organization.
- Compute performance measures such as memory access time, latency, bandwidth, throughput, speedup, hit/miss rates, and resource or power utilization where applicable.
- Interpret the results by evaluating performance improvements, memory bottlenecks, technology trade-offs, and the suitability of the proposed memory hierarchy for each application.

QUESTIONS

1. A large-scale e-commerce platform experiences slow response times during peak shopping periods when thousands of users simultaneously access product catalogs, customer records, and transaction data. Analyze the roles of cache memory, DRAM, and secondary storage in handling these workloads. Propose a memory hierarchy that reduces access latency, improves data locality, and supports high concurrent transaction throughput.
2. A robotic manufacturing system must process camera images, sensor readings, and control instructions with strict real-time deadlines. Evaluate the suitability of SRAM, DRAM, and non-volatile memory for storing frequently accessed control data, sensor information, and program code. Design a memory hierarchy that minimizes latency and ensures predictable memory access for real-time operations.
3. A video-editing workstation experiences delays while processing multiple streams of high-resolution 4K and 8K video. Analyze how cache size, DRAM capacity, memory bandwidth, and storage performance influence video-processing workloads. Recommend a hierarchical memory organization that improves data transfer rates, reduces processing delays, and supports smooth real-time editing.
4. A weather forecasting system performs complex numerical calculations on massive datasets containing atmospheric temperature, pressure, wind, and humidity measurements. Analyze how cache hierarchy, DRAM bandwidth, and large-capacity memory influence the performance of repeated numerical computations. Propose memory-hierarchy enhancements that improve data locality, reduce memory-access bottlenecks, and increase computational throughput.
5. A smart home gateway manages data from cameras, motion sensors, environmental sensors, and connected appliances while operating continuously with limited energy resources. Compare SRAM, DRAM, Flash, and other suitable memory technologies based on latency, capacity, persistence, and power consumption. Design an efficient memory hierarchy that enables rapid access to frequently used data while minimizing energy consumption and maintaining reliable long-term data storage.

Code of Conduct

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Q. No	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professionalism(5)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 25	/ 15	/ 10	/ 5	/ 100

Course Coordinator

Faculty Incharge

1. **A large-scale e-commerce platform experiences slow response times during peak shopping periods when thousands of users simultaneously access product catalogs, customer records, and transaction data. Analyze the roles of cache memory, DRAM, and secondary storage in handling these workloads. Propose a memory hierarchy that reduces access latency, improves data locality, and supports high concurrent transaction throughput.**

Introduction

A large-scale e-commerce platform must handle thousands of users simultaneously accessing product catalogs, customer information, shopping carts, payment details, and transaction records. During peak periods, frequent access to the same data can increase memory latency and reduce system performance. A suitable memory hierarchy using cache memory, DRAM, and secondary storage can reduce access time and improve concurrent transaction throughput.

Role of Cache Memory

Cache memory is the fastest memory available close to the CPU. It stores frequently accessed instructions and data so that the processor does not need to access slower DRAM repeatedly. L1 cache provides the lowest latency, while L2 and L3 caches provide larger storage capacities.

For an e-commerce application, frequently accessed product information, pricing data, session information, and commonly executed database operations can benefit from caching. This improves data locality and reduces repeated memory accesses.

Role of DRAM

DRAM acts as the main working memory of the application servers. Active customer sessions, database buffers, application processes, product information, and transaction data can be stored in DRAM. Although DRAM is slower than cache, it provides much greater capacity.

A larger DRAM capacity allows more frequently used application data to remain in memory, reducing the number of accesses to secondary storage.

Role of Secondary Storage

Secondary storage such as SSDs provides large-capacity and persistent storage. Customer databases, transaction histories, product images, logs, backups, and other permanent data can

be stored on SSDs.

SSDs have lower latency than traditional hard disk drives and are therefore suitable for database storage where frequent read and write operations are required.

Proposed Memory Hierarchy

CPU → L1 Cache → L2 Cache → L3 Cache → DRAM → NVMe SSD → Persistent Storage

Frequently accessed data should remain in higher levels of the hierarchy, while less frequently accessed data can move toward lower levels.

Performance Improvements

- Cache reduces CPU memory-access latency.
- DRAM stores active application and database data.
- SSD provides high-capacity persistent storage.
- Data locality reduces unnecessary memory accesses.
- Database and application caching reduces repeated database queries.
- Multiple server nodes with sufficient DRAM and cache capacity can support high concurrency.

Conclusion

The proposed hierarchy combines the speed of cache, capacity of DRAM, and persistence of secondary storage. By keeping frequently accessed data close to the processor and using SSDs for persistent data, the e-commerce platform can reduce access latency, improve data locality, and support high concurrent transaction throughput.

2. **A robotic manufacturing system must process camera images, sensor readings, and control instructions with strict real-time deadlines. Evaluate the suitability of SRAM, DRAM, and non-volatile memory for storing frequently accessed control data, sensor information, and program code. Design a memory hierarchy that minimizes latency and ensures predictable memory access for real-time operations.**

Introduction

A robotic manufacturing system processes camera images, sensor readings, control instructions, and program code under strict real-time deadlines. In such systems, unpredictable memory delays can affect the timing of robotic operations. Therefore, memory technologies must be selected based on latency, predictability, capacity, and persistence.

SRAM for Frequently Accessed Control Data

SRAM is extremely fast and does not require refresh operations like DRAM. It is suitable for frequently accessed control instructions, real-time variables, processor buffers, and critical sensor information.

The main disadvantage of SRAM is its high cost and lower storage density. Therefore, it should be used for small but critical data.

DRAM for Sensor Information

DRAM provides higher capacity than SRAM and is suitable for storing camera frames, sensor datasets, intermediate processing results, and operating-system data.

However, DRAM has higher latency than SRAM and requires periodic refreshing. For real-time applications, important data should preferably remain in faster memory.

Non-Volatile Memory for Program Code

Non-volatile memory such as Flash can retain information even when power is removed. It is suitable for storing firmware, robot control programs, configuration information, and calibration data.

Program code can be loaded from non-volatile memory into faster memory during system startup.

Proposed Memory Hierarchy

CPU → SRAM/Cache → DRAM → Non-Volatile Flash Memory

- SRAM: Real-time control variables and critical data.
- DRAM: Camera images, sensor readings, and temporary processing data.
- Flash: Firmware, program code, configuration, and persistent data.

Predictable Memory Access

To meet real-time deadlines, critical control data should be locked in SRAM/cache. Frequently accessed instructions should be placed in fast memory to avoid unpredictable delays. DMA can also be used to transfer sensor and camera data directly between devices and memory, reducing CPU overhead.

Data locality should be maintained by keeping frequently used sensor values and control instructions in fast memory.

Advantages

- Very low latency for critical control data.
- Predictable access to real-time information.
- Large sensor and image datasets can be handled using DRAM.
- Program code remains safely stored in non-volatile memory.
- Reduced CPU overhead through efficient data transfers.

Conclusion

A combination of SRAM, DRAM, and non-volatile memory provides an efficient solution for robotic manufacturing. SRAM should handle latency-sensitive control information, DRAM should store high-volume sensor and image data, and Flash should store persistent program code. This hierarchy provides both high performance and predictable real-time operation.

3. A video-editing workstation experiences delays while processing multiple streams of high-resolution 4K and 8K video. Analyze how cache size, DRAM capacity, memory bandwidth, and storage performance influence video-processing workloads. Recommend a hierarchical memory organization that improves data transfer rates, reduces processing delays, and supports smooth real-time editing.

Introduction

Video-editing applications process extremely large amounts of data. 4K and 8K video streams require high memory capacity, high bandwidth, and fast storage. Delays can occur when the processor continuously transfers large video frames between memory and storage. Therefore, cache size, DRAM capacity, memory bandwidth, and storage performance are important factors.

Role of Cache Size

CPU cache stores frequently used instructions and recently accessed video-processing data. A larger cache can improve the probability of finding required data quickly.

L1 cache provides the lowest latency, while L2 and L3 caches provide progressively larger storage capacity. Efficient cache utilization reduces CPU waiting time.

Role of DRAM Capacity

Video editing requires large amounts of working memory for video frames, effects, timelines,

previews, and intermediate results. Insufficient DRAM can cause the operating system to use virtual memory, which increases latency because data must be transferred to and from storage.

Therefore, high-capacity DRAM is important for editing multiple 4K and 8K streams.

Role of Memory Bandwidth

Memory bandwidth determines how much data can be transferred between the processor and main memory per unit of time. High-resolution video requires continuous high-volume data transfers.

Higher bandwidth allows video frames and processing data to move faster, reducing bottlenecks during rendering, effects processing, and playback.

Role of Storage Performance

Fast NVMe SSD storage can significantly improve loading, saving, cache-file creation, and media transfer operations. Slow storage can become a bottleneck when large video files are read or written continuously.

Proposed Memory Hierarchy

CPU → L1/L2/L3 Cache → High-Capacity High-Bandwidth DRAM → NVMe SSD → Long-Term Storage

Frequently processed data should remain in cache and DRAM, while source video and completed projects can be stored on high-speed SSDs and long-term storage.

Optimization Techniques

- Increase DRAM capacity to reduce paging.
- Use high-bandwidth memory configurations.
- Optimize cache locality.
- Use NVMe SSDs for active video projects.
- Separate operating-system and media workloads where possible.
- Use sequential data access for large video files.
- Keep frequently used project data in faster memory.

Conclusion

Smooth 4K and 8K video editing requires a balanced memory hierarchy. Large cache capacity, high-capacity DRAM, high memory bandwidth, and fast NVMe storage work together to reduce data-transfer delays. The proposed hierarchy improves throughput and provides smoother real-time video editing.

4. **A weather forecasting system performs complex numerical calculations on massive datasets containing atmospheric temperature, pressure, wind, and humidity measurements. Analyze how cache hierarchy, DRAM bandwidth, and large-capacity memory influence the performance of repeated numerical computations. Propose memory-hierarchy enhancements that improve data locality, reduce memory-access bottlenecks, and increase computational throughput**

Introduction

Weather forecasting systems perform large-scale numerical calculations using atmospheric datasets containing temperature, pressure, wind speed, humidity, and other measurements. These calculations repeatedly access large datasets. Therefore, memory bandwidth, cache efficiency, data locality, and memory capacity strongly influence computational throughput.

Role of Cache Hierarchy

Weather calculations repeatedly process related data values. If frequently used data is available in the cache, the processor can access it faster than retrieving it from DRAM.

The hierarchy of L1, L2, and L3 caches helps reduce average memory-access latency. Proper data organization can increase cache hit rates.

Role of DRAM Bandwidth

Large weather datasets must continuously move between memory and processors. If DRAM bandwidth is insufficient, processors may remain idle while waiting for data.

High-bandwidth DRAM allows large quantities of temperature, pressure, wind, and humidity data to be transferred efficiently.

Role of Large-Capacity Memory

Weather models can contain extremely large datasets. Sufficient main memory allows more data to remain available without repeatedly transferring information from secondary storage.

If the working dataset exceeds physical memory, paging may occur. Excessive paging increases latency and reduces computational throughput.

Data Locality Optimization

Weather calculations often operate on neighboring values in a grid. Spatial locality can be improved by storing related grid values in contiguous memory locations.

Temporal locality can also be improved by keeping frequently reused intermediate values in cache.

Proposed Memory Hierarchy

Processor → L1 Cache → L2 Cache → L3 Cache → High-Bandwidth DRAM → High-Speed Storage

The processor should perform repeated calculations using data available in the cache whenever possible. Large datasets should remain in DRAM, while permanent weather databases can be stored on high-speed storage.

Performance Enhancements

- Optimize algorithms for cache locality.
- Use blocking or tiling techniques for large matrices.
- Increase DRAM bandwidth.
- Provide sufficient memory capacity.
- Reduce unnecessary data movement.
- Use parallel processing for independent calculations.
- Avoid excessive paging.
- Store frequently reused intermediate results in faster memory.

Conclusion

Weather forecasting is highly dependent on efficient memory access because large datasets are repeatedly processed. A multi-level cache hierarchy combined with high-bandwidth, high-capacity DRAM can reduce memory bottlenecks. Improving locality and minimizing unnecessary data movement increases computational throughput and allows forecasting models to execute more efficiently.

5. **A smart home gateway manages data from cameras, motion sensors, environmental sensors, and connected appliances while operating continuously with limited energy resources. Compare SRAM, DRAM, Flash, and other suitable memory technologies based on latency, capacity, persistence, and power consumption. Design an efficient memory hierarchy that enables rapid access to frequently used data while minimizing energy consumption and maintaining reliable long-term data storage.**

Introduction

A smart home gateway continuously manages data from cameras, motion sensors, environmental sensors, and connected appliances. Since the gateway operates continuously with limited energy resources, the memory system must provide fast access while minimizing power consumption. It must also preserve important information even after power is removed.

SRAM

SRAM provides very low latency and is suitable for frequently accessed control information. It can store active sensor states, device-control variables, and critical gateway operations. However, SRAM has high cost per bit and lower capacity, so it should be used only for critical and frequently accessed data.

DRAM

DRAM provides greater capacity than SRAM and can store active application data, camera buffers, sensor information, and temporary processing data. However, DRAM consumes power because it requires periodic refresh operations. Therefore, its capacity should be selected according to the gateway workload.

Flash Memory

Flash is non-volatile and retains information without continuous power. It is suitable for firmware, configuration files, event logs, device information, and long-term data. Flash has higher access latency than SRAM and DRAM, so it should not normally be used for highly latency-sensitive operations.

Other Suitable Technologies

Technologies such as MRAM can provide non-volatile storage with fast access and good endurance. Depending on system requirements and cost, MRAM can be useful for frequently updated persistent information.

Proposed Memory Hierarchy

CPU → SRAM/Cache → DRAM → Flash/MRAM → Long-Term Storage

- SRAM/Cache: Frequently accessed control data.
- DRAM: Active sensor and camera processing data.
- MRAM/Flash: Persistent configuration, logs, and firmware.
- Long-term storage: Historical sensor and camera data.

Power Optimization

The gateway can reduce energy consumption by keeping only frequently accessed information in SRAM and DRAM. Infrequently used data can be moved to non-volatile memory.

Power-saving techniques such as memory sleep modes, efficient data buffering, reduced unnecessary writes, and intelligent caching can further reduce energy consumption.

Reliability

Important configuration data and firmware should be stored in non-volatile memory. Data redundancy and error detection can be used for important information. Logs can be periodically transferred to persistent storage rather than continuously keeping all data in high-power memory.

Conclusion

An efficient smart-home gateway should combine SRAM, DRAM, and non-volatile memory. SRAM provides rapid access to critical data, DRAM handles active workloads, and Flash or MRAM provides persistent storage. This hierarchy balances latency, capacity, persistence, reliability, and energy consumption while supporting continuous smart-home operation.